CS 532
18 Quizzes — Organized by Lecture

Exam-reference edition

Each original question and subquestion, with numbered red steps, substitutions, equations, verified answers, and completed diagrams.

Scope: The 18 quizzes supplied in the source guide, grouped by primary lecture match (Lectures 1–6). The original relevance scores and duplicates are preserved; two additional quizzes mentioned in the study outline were not supplied and are not invented.

Conventions: Image points are homogeneous unless otherwise indicated. A virtual image plane lies at camera $z = f$ for pinhole derivations. Camera/world transformation direction is always specified explicitly.

How to use during an exam: Use the lecture sections, contents page, quiz title, or formula locator. In every problem, the question is in black, the complete method is red, and important caveats are gray.

Source fidelity: The question statements follow the uploaded old quiz PDFs (edited slightly for typography/clarity but never changing numerical givens). Where the supplied Fall 2025 answer key uses ambiguous or inaccurate notation, a clearly labeled check follows it. Additional example calculations appear only when labeled *illustrative*.

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Lecture organization and score legend

Primary lecture match: Every quiz appears once, under its principal lecture. Secondary lecture overlaps are mentioned in section study notes.

Relevance grades: A = highest relevance; B = strong relevance; C = partial relevance; D = outside the six-lecture scope. The numerical scores shown in quiz headings are *relevance scores, not grades on the worked solutions.*

Count clarification: The supplied outline says "20 quizzes," but the six tables list 18 entries, matching the 18 quizzes in this source. This edition therefore includes 18 historical quizzes plus seven clearly labeled original practice problems.

Fast formula locator

Concept	Core relation / worked quizzes
Epipolar geometry	$\ell' = Fx$, $x'^T Fx = 0$, $e'^T F = 0$; Lecture 4
Stereo geometry	$Z = bf/d$, $\Delta Z \approx -bf\,\Delta d/d^2$; Lecture 4
Homography/RANSAC	$x' \sim Hx$, $N \geq \ln(1-p)/\ln(1-w^s)$; Lecture 3
Projective primitives	$\ell = a \times b$, $c^T \ell = 0$; Lecture 2
KLT brightness	$I_x u + I_y v + I_t = 0$; Lecture 5
Camera geometry	$x \sim K[R \mid t]X$, $f_x = (W/2)/\tan(\theta/2)$; Lecture 3
SfM	Gauge, plane at infinity, incremental reconstruction, bundle adjustment; Lecture 6
Harris	$M = \sum w \nabla I \nabla I^T$, $R = \det M - k(\operatorname{tr} M)^2$; Lecture 5
Rigid geometry	Pose composition, plane normals, $R_z(90°)$; Lecture 1

Key

unit warning: In $Z = bf/d$, b and Z have the same physical units; f and d use the same image-coordinate units (usually pixels). Distinguish *exact* depth error from the *first-order* Taylor approximation.

Common symbol warning: Fx and $F^T x'$ are *line coefficient vectors*, not generally pixel coordinates. A line $\ell = (a, b, c)^T$ means $au + bv + c = 0$ for image coordinates (u, v).

Lecture 1 — Foundations and 3D Geometry

Lecture study note. Supplementary practice: camera-relative poses, 3D planes, depth-map backprojection and surface normals. Also relevant to Lectures 3–4.

Relevance scores in this section are from the original quiz-matching rubric, not solution grades.

Fall 2025 Quiz 1 — Probability and rigid camera poses **69/100**

Source: `cs532f25_quiz_1_solution.pdf`
Lecture match: Lectures 1–4; RANSAC probability analogy

Question 1. A parlay succeeds only if all three independent fair coin flips succeed. What is the chance of winning one parlay?

Step 1. Each fair flip succeeds with probability 1/2. The three flip outcomes are independent.

Step 2. Multiply probabilities for the joint event: $P(\text{all 3}) = (1/2)^3 = 1/8$.

Step 3. $\boxed{P(\text{win}) = 1/8 = 12.5\%}$. This is the same product rule used for all-inlier RANSAC samples.

Question 2. You attempt one three-coin-flip parlay each day for seven independent days. What is the probability of winning at least one?

Step 1. One parlay win probability is $q = 1/8$. Its loss probability is $1 - q = 7/8$.

Step 2. All seven attempts fail with probability $P(0 \text{ wins}) = (7/8)^7$.

Step 3. Use the complement of no wins:

$$P(\geq 1 \text{ win}) = 1 - (7/8)^7 = 1 - 823543/2097152.$$

Step 4. Therefore $\boxed{P(\geq 1 \text{ win}) \approx 0.607304 \approx 60.73\%}$.

Question 3. Two static cameras concurrently estimate the 3D rigid pose of the same object, as $[R_1, t_1]$ and $[R_2, t_2]$ in their own camera frames. Can one recover the relative camera pose? Explain and derive a transformation.

Step 1. Define ${}^{C_i}T_O = \begin{bmatrix} R_i & t_i \\ 0 & 1 \end{bmatrix}$, which maps an object-frame homogeneous point X_O into camera frame C_i: $X_{C_i} = {}^{C_i}T_O X_O$. Both poses refer to the *same object at the same time.*

Step 2. Express the object coordinates from camera-2 coordinates: $X_O = ({}^{C_2}T_O)^{-1} X_{C_2}$.

Step 3. Insert into camera-1 expression: $X_{C_1} = {}^{C_1}T_O ({}^{C_2}T_O)^{-1} X_{C_2}$.

Step 4. Thus $\boxed{{}^{C_1}T_{C_2} = {}^{C_1}T_O ({}^{C_2}T_O)^{-1}}$. The inverse gives ${}^{C_2}T_{C_1}$.

Step 5. Expand if asked: inverse of $[R_2, t_2]$ is $[R_2^T, -R_2^T t_2]$; therefore $R_{12} = R_1 R_2^T$, $t_{12} = t_1 - R_1 R_2^T t_2$.

Exam check / qualification. The uploaded key's superscript/subscript arrows are inconsistent in places. The formula above is unambiguous because its mapping direction is defined explicitly. If the poses are only known up to unknown scale, relative metric translation is not determined absolutely.

Question 4. Can one determine the 3D plane passing through both camera centers and the tracked object's center? Derive it.

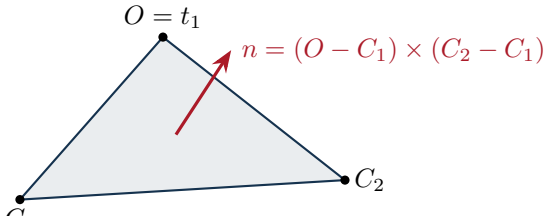
Step 1. Use camera-1 coordinates as the world frame: $C_1 = (0, 0, 0)^T$. The object-frame origin is observed at $O = t_1$.

Step 2. Find camera-2 center C_2 as the translation part of the transform ${}^{C_1}T_{C_2}$ from Question 3.

Step 3. Two vectors lying in the plane are $a = O - C_1$ and $b = C_2 - C_1$. Take the cross product to find a plane normal: $\boxed{n = a \times b}$.

Step 4. The plane equation is $n^T(X - C_1) = 0$ or $n^T X + d = 0$ with $d = -n^T C_1$. Because $C_1 = 0$, $\boxed{n^T X = 0}$.

Step 5. Require $n \neq 0$, i.e. the three points are noncollinear. If cameras and object center lie on one line, infinitely many planes contain them. Draw a triangle joining C_1, C_2, O with a perpendicular arrow n.



Plane through three noncollinear points (perspective sketch)

Fall 2017 Quiz 9 — Depth maps and tetrahedron normals **55/100**

Source: `cs532_f17_quiz9.pdf`
Lecture match: Lectures 1–4 transfer; normals/point-cloud topics are only partly in scope

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Problem 1. Explain how to convert two depth maps into a single 3D point cloud in a common reference frame. What information is required? Make a sketch.

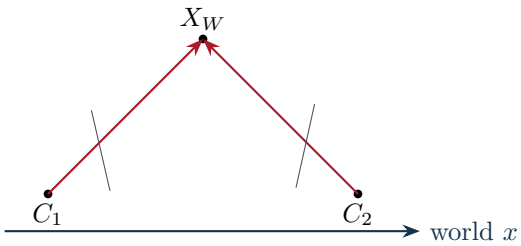
Step 1. For camera $i \in \{1, 2\}$ obtain intrinsic matrix K_i, per-pixel depth $Z_i(u, v)$, and a known camera-to-world rigid transform ${}^{W}T_{C_i} = \begin{bmatrix} R_i & t_i \\ 0 & 1 \end{bmatrix}$. First correct lens distortion if depth is reported in distorted coordinates.

Step 2. For pixel $p_i = (u, v, 1)^T$, compute normalized camera ray $r_i = K_i^{-1} p_i$, whose last component is 1 under standard intrinsics. If depth means camera-axis Z, backproject: $\boxed{X_{C_i} = Z_i(u, v)\, K_i^{-1} p_i}$.

Step 3. For standard intrinsics with zero skew this becomes $X_{C_i} = ((u - c_x)Z/f_x, (v - c_y)Z/f_y, Z)^T$. Skip missing, zero, invalid, or occluded depth pixels.

Step 4. Map every 3D point into the *same* world frame: $\boxed{X_W = R_i X_{C_i} + t_i}$. Combine both transformed sets; optional voxel fusion/outlier removal prevents duplicates or inconsistent observations.

Step 5. If depth means *Euclidean range along ray*, use $X_{C_i} = \rho_i r_i / \|r_i\|$, not $Z_i r_i$. The question needs a depth convention, the two sets of intrinsics, and the relative/world poses.



Both backprojected clouds must be expressed in the *same* frame.

Problem 2(a). A tetrahedron has vertices $v_1 = (1, 1, 0)$, $v_2 = (1, 2, 0)$, $v_3 = (1, 1, 1)$, $v_4 = (2, 1, 0)$. Find one *outward* surface normal per face, and sketch the result.

Step 1. Choose the four triangular faces: $(1, 2, 3)$, $(1, 2, 4)$, $(1, 3, 4)$, and $(2, 3, 4)$. For vertices i, j, k, calculate candidate $n_{ijk} = (v_j - v_i) \times (v_k - v_i)$.

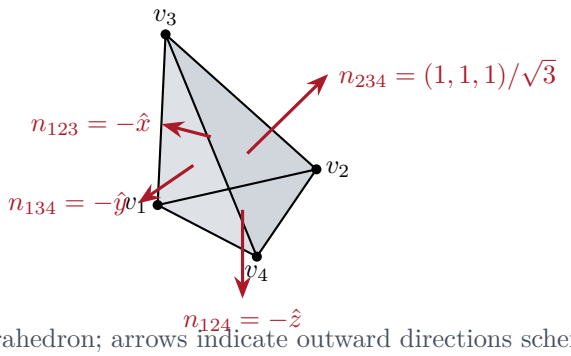
Step 2. Face 123: $v_2 - v_1 = (0, 1, 0)$ and $v_3 - v_1 = (0, 0, 1)$; cross product gives $(1, 0, 0)$. Since the opposite vertex v_4 has $x = 2$, interior points face +x, so outward normal is $\boxed{n_{123} = (-1, 0, 0)}$.

Step 3. Face 124: $v_2 - v_1 = (0, 1, 0)$, $v_4 - v_1 = (1, 0, 0)$; cross product is $(0, 0, -1)$, pointing away from opposite v_3 above $z = 0$. Thus $\boxed{n_{124} = (0, 0, -1)}$.

Step 4. Face 134: $v_3 - v_1 = (0, 0, 1)$, $v_4 - v_1 = (1, 0, 0)$; cross product $(0, 1, 0)$ points toward opposite v_2 with $y = 2$. Flip: $\boxed{n_{134} = (0, -1, 0)}$.

Step 5. Face 234: $v_3 - v_2 = (0, -1, 1)$ and $v_4 - v_2 = (1, -1, 0)$. Cross product $(1, 1, 1)$, which points away from opposite vertex v_1 across plane $x + y + z = 3$. Thus $\boxed{n_{234} = (1, 1, 1)/\sqrt{3}}$ for a unit normal.

Step 6. Sketch triangular faces forming the tetrahedron and draw the four arrows perpendicular to their face planes, directed away from its interior. The first three have unit magnitude already; the fourth needs division by $\sqrt{3}$ for unit length.



Projected 3D tetrahedron; arrows indicate outward directions schematically

Problem 2(b). Rotate the complete tetrahedron by $+90°$ about the global Z axis. Find the modified face normals.

Step 1. For an *active*, right-handed $+90°$ rotation,

$$R_z(90°) = \begin{bmatrix} \cos 90° & -\sin 90° & 0 \\ \sin 90° & \cos 90° & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Step 2. Pure rigid rotations rotate normals by the same matrix: $n'_{ijk} = R_z n_{ijk}$. (General nonrigid affine transforms use an inverse-transpose rule instead.)

Step 3. Compute each: $R_z(-1,0,0)^T = (0,-1,0)^T$; $R_z(0,0,-1)^T = (0,0,-1)^T$; $R_z(0,-1,0)^T = (1,0,0)^T$.

Step 4. For the slanted face, $R_z(1,1,1)^T/\sqrt{3} = (-1,1,1)^T/\sqrt{3}$.

Step 5. Collect the results:

$$\boxed{n'_{123} = (0,-1,0), \quad n'_{124} = (0,0,-1),}$$

$$\boxed{n'_{134} = (1,0,0), \quad n'_{234} = (-1,1,1)/\sqrt{3}.}$$

Lecture 2 — Projective Geometry and Linear Algebra

Lecture study note. Coverage gap: advanced rotation representations, quaternions and transformation hierarchies are not directly tested by these historical quizzes. Also relevant to Lecture 1.

Relevance scores in this section are from the original quiz-matching rubric, not solution grades.

Spring 2020 Quiz 1 — Projective points and camera projection **96/100**

Source: `cs532s20_quiz1.pdf`
Lecture match: Lectures 1–2

Problem 1. For projective-plane points a, b, c, use only dot and cross products to test whether they are collinear. In what case does the test fail?

Step 1. Recall that the homogeneous *line through two distinct points* is their cross product: $\ell = a \times b$.

Step 2. A third point c lies on ℓ exactly when its dot product with ℓ is zero. Thus test $\boxed{c^T(a \times b) = 0}$. This equals the determinant of the matrix with columns a, b, c.

Step 3. Check degeneracy: if $a \sim b$ (they represent the same projective point), then $a \times b = 0$, and the test returns zero for *every* c, including noncollinear ones. Use a different pair of distinct representatives if possible. The zero homogeneous vector is invalid as a point.

Problem 2(a–b). Let $X = (X, Y, Z, W)^T$ be a homogeneous 3D point and $x \sim MX$ a homogeneous 2D image point. Find (a) the dimensions of M; (b) explicit inhomogeneous image coordinates from the entries m_{ij} and X.

Step 1. A 3D homogeneous point has four components; a 2D homogeneous image point has three. Therefore $\boxed{M \in \mathbb{R}^{3\times 4}}$ and Mx is *not* the mapping; the correct product is $MX \in \mathbb{R}^{3\times 1}$.

Step 2. Write $M = [m_{ij}]$ and compute $q = MX = (q_1, q_2, q_3)^T$ with $q_i = m_{i1}X + m_{i2}Y + m_{i3}Z + m_{i4}W$.

Step 3. Convert homogeneous to inhomogeneous coordinates by dividing by the third component:

$$\boxed{u = \frac{m_{11}X + m_{12}Y + m_{13}Z + m_{14}W}{m_{31}X + m_{32}Y + m_{33}Z + m_{34}W}}, \quad \boxed{v = \frac{m_{21}X + m_{22}Y + m_{23}Z + m_{24}W}{m_{31}X + m_{32}Y + m_{33}Z + m_{34}W}}.$$

For ordinary 3D coordinates set $W = 1$; require $q_3 \neq 0$ to get a finite pixel.

Problem 3(a–b). Let ℓ_1, ℓ_2 be two distinct vertical lines each at distance $d > 0$ from the 2D origin, and ℓ_3 the horizontal line through the origin. Give their homogeneous line vectors and count the possible pairwise intersections.

Step 1. Choose $\ell_1 : x = d$ and $\ell_2 : x = -d$; as $ax + by + c = 0$, their vectors are $\boxed{\ell_1 = (1, 0, -d)^T}$, $\boxed{\ell_2 = (1, 0, d)^T}$.

Step 2. For the horizontal line $y = 0$, use $\boxed{\ell_3 = (0, 1, 0)^T}$. Line vectors are homogeneous, so nonzero multiples are equivalent.

Step 3. Intersect ℓ_1, ℓ_3: $\ell_1 \times \ell_3 = (d, 0, 1)^T \Rightarrow \boxed{(d, 0)}$. Intersect ℓ_2, ℓ_3: $\ell_2 \times \ell_3 = (-d, 0, 1)^T \Rightarrow \boxed{(-d, 0)}$.

Step 4. Intersect the *parallel* vertical lines: $\ell_1 \times \ell_2 = (0, -2d, 0)^T \sim (0, 1, 0)^T$, a *point at infinity*, not a finite 2D coordinate.

Step 5. Thus $\boxed{\text{3 projective intersections}}$ among the 3 line pairs, of which $\boxed{\text{2 are finite}}$. Say both numbers to clarify the geometric convention.

Lecture 3 — Camera Projection, Homographies, and RANSAC

Lecture study note. Spring 2019 Quiz 2 and Spring 2020 Quiz 3 are near-duplicates. Coverage gaps: numerical DLT, SVD, normalized homography estimation and image warping. The supplement provides DLT/SVD practice. Secondary overlaps: Lectures 1–2.

Relevance scores in this section are from the original quiz-matching rubric, not solution grades.

Spring 2019 Quiz 3 — Pentagon, homography, RANSAC **97/100**

Source: `cs532s19-quiz3.pdf`
Lecture match: Lecture 3, slides 34–55

Problem 1(a–d). Two images show only the same *irregular five-sided polygon*, related by an unknown homography. All five corners are detected in each image. You create every possible putative corner correspondence. (a) How many? (b) What inlier rate? (c) How many RANSAC iterations are needed to reliably estimate the homography? (d) What changes for a regular/equilateral pentagon?

Step 1. There are 5 features in image A and 5 in image B. Exhaustive matching forms all ordered cross-image pairs, so $\boxed{5 \times 5 = 25}$ putative correspondences.

Step 2. Assume one unique correct correspondence for each of the 5 image-A corners. Then 5 of the 25 pairs are inliers, so $\boxed{w = 5/25 = 0.2}$ and outlier rate $e = 0.8$.

Step 3. The minimal homography solver requires $s = 4$ nondegenerate correspondences. The usual RANSAC independent-sample approximation sets

$$1 - (1 - w^s)^N \geq p \quad \Rightarrow \quad N \geq \frac{\ln(1-p)}{\ln(1-w^s)}.$$

Step 4. For the *illustrative* common confidence $p = 0.99$, $w^4 = 0.2^4 = 0.0016$, hence

$$\boxed{N_{\text{approx}} = \left\lceil \frac{\ln(0.01)}{\ln(0.9984)} \right\rceil = 2876}.$$

The quiz does not specify a confidence; the general formula is the source-supported answer. Sampling all putative matches independently also ignores the one-to-one matching constraint, so these iteration counts describe a simplified model.

Step 5. Because there are only 25 candidate pairs and each RANSAC *sample selects 4 distinct pairs*, the exact probability of four inliers in a uniform draw without replacement is smaller:

$$q_{\text{exact}} = \frac{\binom{5}{4}}{\binom{25}{4}} = \frac{5}{12650} \approx 0.00039526.$$

If *each trial samples afresh* and we desire $p = 0.99$,

$$\boxed{N_{\text{finite}} = \left\lceil \frac{\ln(.01)}{\ln(1 - 5/12650)} \right\rceil = 11649}.$$

This is an optional exact finite-population refinement; the standard lecture approximation uses w^s.

Step 6. For a regular pentagon, rotational/reflection symmetry makes the corner identities ambiguous: several match permutations may fit geometrically similar configurations. Without texture/orientation/other cues, a unique physical correspondence is less secure. Explain that good geometric fits need not identify the *intended* corner labeling.

Exam check / qualification. The irregular-pentagon problem still requires nondegenerate 4-tuples (e.g. no three source points collinear). The exact without-replacement refinement above does not imply that *every* four-inlier subset yields a uniquely acceptable model under all degeneracies.

Spring 2019 Quiz 2 — FOV, calibration, homogeneous scaling **93/100**

Source: `cs532s19_quiz2-2.pdf`
Lecture match: Lectures 1–3

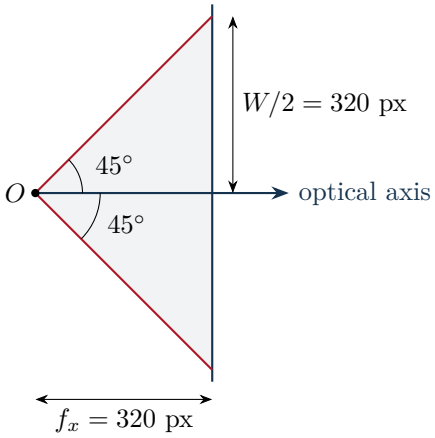
Problem 1. A digital camera has image resolution 640×480 pixels and horizontal field of view $90°$. Determine the focal length in pixels and draw a sketch.

Step 1. Take a horizontal cross-section through the optical axis. The full horizontal FOV is $90°$, so each edge ray is $45°$ from the optical axis. The image half-width is $W/2 = 320$ pixels.

Step 2. In the pinhole triangle, $\tan(\theta/2) = (W/2)/f_x$. Rearrange: $f_x = (W/2)/\tan(\theta/2)$.

Step 3. Substitute $W = 640$, $\theta = 90°$:

$$\boxed{f_x = \frac{320}{\tan 45°} = 320 \text{ px}}.$$



Problem 2. Write the intrinsic calibration matrix K for the camera.

Step 1. Use standard intrinsic form $K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$.

Step 2. The horizontal FOV determines $f_x = 320$ px only. To fill the other terms, *assume* square pixels ($f_y = f_x$), zero skew ($s = 0$), and centered principal point ($(c_x, c_y) = (320, 240)$).

Step 3. Under those common assumptions,

$$\boxed{K = \begin{bmatrix} 320 & 0 & 320 \\ 0 & 320 & 240 \\ 0 & 0 & 1 \end{bmatrix}}.$$

Without them, f_y, s, c_x, c_y cannot all be inferred from the given horizontal FOV alone.

Problem 3(a–b). In $x = K[R \mid t]X$, give dimensions of every matrix/vector. If the right-hand side is multiplied by 2 to obtain $x' = 2K[R \mid t]X$, how are x' and x related?

Step 1. $K : 3 \times 3$, $R : 3 \times 3$, $t : 3 \times 1$, $[R \mid t] : 3 \times 4$, $X : 4 \times 1$, $x : 3 \times 1$; total projection matrix $P = K[R \mid t]$ is 3×4.

Step 2. Homogeneous images are defined up to nonzero scale: $x' = (2q_1, 2q_2, 2q_3)^T$ vs $x = (q_1, q_2, q_3)^T$.

Step 3. Dehomogenize: $u' = 2q_1/(2q_3) = q_1/q_3 = u$ and $v' = 2q_2/(2q_3) = v$. So $\boxed{x' \sim x}$: identical physical image points.

Problem 4. What are the 3D coordinates of the image principal point in the camera coordinate system?

Step 1. The principal point is the intersection of the camera optical axis with the image plane. In a virtual plane $z = f_{\text{physical}}$, the optical axis is $X = Y = 0$.

Step 2. Therefore its camera-frame position is $\boxed{(0, 0, f_{\text{physical}})}$. If the image-plane coordinate units are *chosen as pixel-equivalent units*, one may write $(0, 0, 320)$ using the computed focal length in pixels.

Step 3. Distinguish this 3D position from the image-array pixel coordinates $\boxed{(c_x, c_y) = (320, 240)}$ under the centered-principal-point assumption. Physical f in meters/mm is not determined by pixel focal length alone without pixel pitch.

Spring 2020 Quiz 3 — Duplicate camera model **93/100**

Source: `cs532s20_quiz3.pdf`
Lecture match: Lectures 1–3

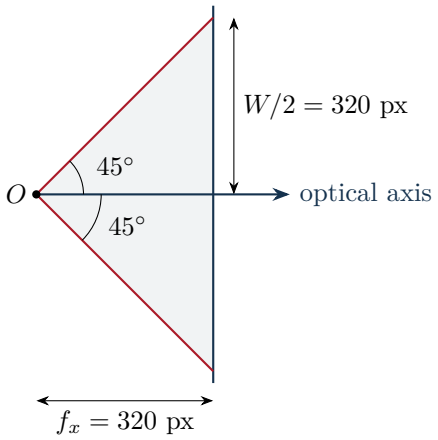
Near duplicate: This PDF repeats rank 10's four camera-calibration questions; independent fully worked answers follow.

Problem 1. A camera is 640 **pixels wide by** 480 **tall with horizontal FOV** 90°**. Find pixel focal length and sketch the rays.**

Step 1. Half-angle is $90°/2 = 45°$, and half-width is $640/2 = 320$ pixels. Draw the camera center, optical axis and the rays to the two horizontal ends of the image plane.

Step 2. $\tan 45° = (320\,\text{px})/f_x$.

Step 3. Since $\tan 45° = 1$, $\boxed{f_x = 320\,\text{px}}$. The half-angle/FOV diagram on the preceding quiz applies.



Problem 2. Write the camera intrinsic matrix K.

Step 1. Use $K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$. The supplied FOV fixes only f_x.

Step 2. Explicitly assume square pixels $f_y = f_x = 320$, zero skew $s = 0$, and principal point at $(320, 240)$.

Step 3. Then $\boxed{K = \begin{bmatrix} 320 & 0 & 320 \\ 0 & 320 & 240 \\ 0 & 0 & 1 \end{bmatrix}}$.

Problem 3(a–b). Give dimensions in $x = K[R \mid t]X$ and explain what doubling the full homogeneous right-hand side changes.

Step 1. Identify $K \in \mathbb{R}^{3\times3}$, $R \in \mathbb{R}^{3\times3}$, $t \in \mathbb{R}^{3\times1}$, $[R \mid t] \in \mathbb{R}^{3\times4}$, $X \in \mathbb{R}^{4\times1}$, $x \in \mathbb{R}^{3\times1}$.

Step 2. For projected $x = (a, b, c)^T$, the doubled homogeneous vector $x' = 2x = (2a, 2b, 2c)^T$.

Step 3. The dehomogenized coordinates are the same: $(a/c, b/c) = (2a/2c, 2b/2c)$. Answer: $\boxed{\text{no image location change}}$.

Problem 4. Find the 3D coordinates of the principal point in the camera reference frame.

Step 1. Set the optical-axis ray to $\{(0, 0, Z) : Z > 0\}$; intersect with virtual image plane $z = f_{\text{physical}}$.

Step 2. The point is $\boxed{(0, 0, f_{\text{physical}})}$ in camera length units; in pixel-normalized plane units $f = 320$ gives the coordinate triple $(0, 0, 320)$.

Step 3. If the question instead asks for image-array pixel coordinates, use $(c_x, c_y) = (320, 240)$ under the stated centering assumption.

Spring 2020 Quiz 2 — Pinhole ray and cross product **90/100**

Source: `cs532s20_quiz2.pdf`
Lecture match: Lectures 2–3

Problem 1. Let $A = (X, Y, Z)$ be a camera-frame 3D point. Its viewing ray passes through image point a on the image plane. Given $f = Z/10$, determine the 3D camera-frame coordinates of a; sketch.

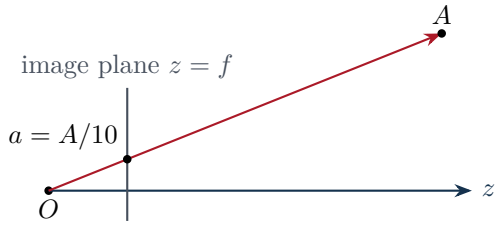
Step 1. For a virtual pinhole image plane at $z = f$, the 3D point and pixel lie on the same ray from the origin. Similar triangles give $x_a/f = X/Z$ and $y_a/f = Y/Z$.

Step 2. Thus $a = (fX/Z, fY/Z, f)$.

Step 3. Substitute $f = Z/10$, assuming $Z \neq 0$:

$$\boxed{a = (X/10, Y/10, Z/10) = A/10}.$$

Step 4. Draw origin O, the point A, the intermediate image plane $z = f$, and a on the segment of ray OA (for $Z > 0$ and $f = Z/10$). The vectors Oa and OA point in the same direction.



Problem 2. Develop the expression for $A \times a$.

Step 1. Set $A = (X, Y, Z)^T$ and $a = (X/10, Y/10, Z/10)^T$.

Step 2. Use the cross-product determinant:

$$A \times a = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ X & Y & Z \\ X/10 & Y/10 & Z/10 \end{vmatrix}.$$

Step 3. Expand components: $(YZ/10 - ZY/10,\ ZX/10 - XZ/10,\ XY/10 - YX/10)$.

Step 4. Every component is zero, so $\boxed{A \times a = (0, 0, 0)^T}$. Geometrically, collinear vectors have zero cross product.

Older Quiz 9 — Pinhole ray intersects a 3D plane **88/100**

Source: `cs532f15_quiz9.pdf`
Lecture match: Lectures 2–3; source PDF header: Spring 2019 Quiz 9

Problem 1. A 3D plane has homogeneous parameters $\pi = (A, B, C, D)^T$, i.e. $AX + BY + CZ + D = 0$. A pinhole camera has center at the 3D origin, orientation I, focal length f, and observes pixel image-plane coordinates (x', y'). Find the 3D depth of the intersection of the viewing ray with the plane. Draw a sketch.

Step 1. The ray begins at camera origin and passes through (x', y', f) on the virtual image plane. Its parametric form is $r(\lambda) = \lambda(x', y', f) = (\lambda x', \lambda y', \lambda f)$.

Step 2. Using pinhole geometry, $x'/f = X/Z$, $y'/f = Y/Z$, hence $X = (x'/f)Z$ and $Y = (y'/f)Z$.

Step 3. Substitute into plane equation:

$$A\frac{x'Z}{f} + B\frac{y'Z}{f} + CZ + D = 0.$$

Step 4. Factor out Z: $Z(Ax'/f + By'/f + C) = -D$.

Step 5. Solve the optical-axis depth:

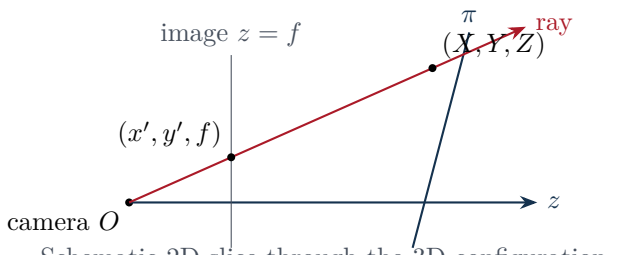
$$\boxed{Z = -\frac{Df}{Ax' + By' + Cf}}.$$

Step 6. Recover the complete intersection point, if requested:

$$\boxed{X = \frac{x'Z}{f}, \quad Y = \frac{y'Z}{f}, \quad Z = -\frac{Df}{Ax' + By' + Cf}}.$$

Equivalently $\lambda = -D/(Ax' + By' + Cf)$ and $r(\lambda)$ gives all three coordinates.

Step 7. Check exceptions: if $Ax' + By' + Cf = 0$ and $D \neq 0$, the viewing ray is parallel to the plane (no finite intersection). If both equal zero, the entire ray lies in the plane (no unique depth). A visible front-of-camera intersection normally requires $Z > 0$.



Schematic 2D slice through the 3D configuration.

Exam check / qualification. The formula gives Z (depth along optical axis), not necessarily Euclidean camera-to-point range $\sqrt{X^2 + Y^2 + Z^2}$.

Lecture 4 — Two-View Geometry and Stereo

Lecture study note. Most heavily represented lecture. Duplicate pairs: 2015 Quiz 4 / Spring 2019 Quiz 4, and Spring 2019 Quiz 5 / Spring 2020 Quiz 5. Coverage gaps: essential-matrix decomposition, normalized eight-point estimation, triangulation and stereo matching costs. See supplemental examples.

Relevance scores in this section are from the original quiz-matching rubric, not solution grades.

Older Quiz 4 — Epipolar line and constraint **100/100**

Source: `cs532f15_quiz4.pdf`
Lecture match: Lecture 4, slides 18–21

Problem 1. **Let $x_A = (x, y, w)^T$ be a point in image A and $F_{AB} = [f_{ij}]$ be the fundamental matrix from image A to B. Write the epipolar-line equation in B, using both matrix entries and the row vectors $\mathbf{f}_1^T, \mathbf{f}_2^T, \mathbf{f}_3^T$. Make an illustrative sketch.**

Given / relevant facts: $F_{AB} = \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}$; homogeneous line $\ell_B = (a, b, c)^T$.

Step 1. Start from the fundamental-matrix point-to-line relation: $\ell_B = F_{AB} x_A$. This is a line, not the matched pixel itself.

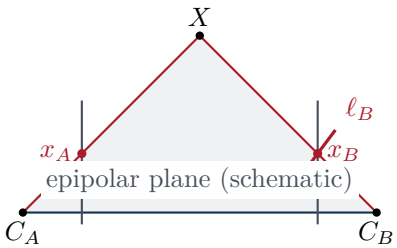
Step 2. Perform the matrix multiplication:

$$\ell_B = \begin{bmatrix} f_{11}x + f_{12}y + f_{13}w \\ f_{21}x + f_{22}y + f_{23}w \\ f_{31}x + f_{32}y + f_{33}w \end{bmatrix} = \begin{bmatrix} \mathbf{f}_1^T x_A \\ \mathbf{f}_2^T x_A \\ \mathbf{f}_3^T x_A \end{bmatrix}.$$

Step 3. A corresponding pixel $x_B = (u_B, v_B, w_B)^T$ must lie on this line, hence $x_B^T \ell_B = 0$. Expand:

$$(f_{11}x + f_{12}y + f_{13}w)u_B + (f_{21}x + f_{22}y + f_{23}w)v_B + (f_{31}x + f_{32}y + f_{33}w)w_B = 0.$$

Step 4. If the point is finite, set $w_B = 1$; plot $au_B + bv_B + c = 0$. For $b \neq 0$, $v_B = -(a/b)u_B - c/b$. Draw a line in image B through the unknown match; its exact location depends on the supplied F_{AB} and x_A.



Exam check / qualification. The 2D drawing is a cross-section of the epipolar plane. The actual epipolar line is the intersection of that 3D plane with image plane B; its apparent slant in the sketch is illustrative.

Problem 2. **Write the explicit epipolar constraint $x_B^T F_{AB} x_A = 0$ for $x_A = (1, 0, 1)^T$ and $x_B = (0, 1, 1)^T$. Rewrite it using row-vector dot products.**

Step 1. Apply F_{AB} to x_A: only its first and third columns contribute:

$$F_{AB} x_A = (f_{11} + f_{13},\ f_{21} + f_{23},\ f_{31} + f_{33})^T.$$

Step 2. Left-multiply by $x_B^T = [0\ 1\ 1]$:

$$x_B^T F_{AB} x_A = 0(f_{11} + f_{13}) + (f_{21} + f_{23}) + (f_{31} + f_{33}).$$

Step 3. Enforce incidence:

$$\boxed{f_{21} + f_{23} + f_{31} + f_{33} = 0}.$$

Step 4. Rewrite using the supplied row vectors: $x_B^T F x_A = (\mathbf{f}_2^T + \mathbf{f}_3^T)x_A = \mathbf{f}_2 \cdot x_A + \mathbf{f}_3 \cdot x_A = 0$.

Exam check / qualification. The quiz provides symbolic F, so no individual numeric f_{ij} values can be solved; the requested result is a constraint on the entries.

Spring 2019 Quiz 4 — Duplicate epipolar retest **100/100**

Source: `cs532s19_quiz4.pdf`
Lecture match: Lecture 4, slides 18–21

Near duplicate: The questions in this uploaded PDF are effectively identical to rank 1. Full steps are repeated here so either quiz can be studied independently.

Problem 1. Given $x_A = (x, y, w)^T$ and the rows $\mathbf{f}_1^T, \mathbf{f}_2^T, \mathbf{f}_3^T$ of F_{AB}, express its epipolar line in image B and make a sketch.

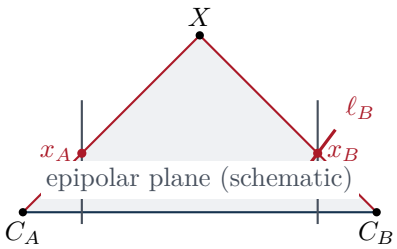
Step 1. Choose the correct direction: F_{AB} maps a *point in A* to a *line in B*; compute $\ell_B = F_{AB} x_A$.

Step 2. Expand the three line coefficients:

$$a = f_{11}x + f_{12}y + f_{13}w, \quad b = f_{21}x + f_{22}y + f_{23}w, \quad c = f_{31}x + f_{32}y + f_{33}w.$$

Step 3. The line is $\boxed{au_B + bv_B + cw_B = 0}$, or $\boxed{au_B + bv_B + c = 0}$ for finite pixels. Equivalent row form: $\ell_B = (\mathbf{f}_1^T x_A, \mathbf{f}_2^T x_A, \mathbf{f}_3^T x_A)^T$.

Step 4. Sketch two camera centers joined by a baseline, their image planes, a common 3D point and both rays; label the line in image B that contains the possible match. The same diagram as on the preceding page applies.



Exam check / qualification. The 2D drawing is a cross-section of the epipolar plane. The actual epipolar line is the intersection of that 3D plane with image plane B; its apparent slant in the sketch is illustrative.

Problem 2. For $x_A = (1, 0, 1)^T$ and $x_B = (0, 1, 1)^T$, expand $x_B^T F_{AB} x_A = 0$ and express it in terms of row vectors.

Step 1. Compute $F_{AB} x_A = (f_{11} + f_{13}, f_{21} + f_{23}, f_{31} + f_{33})^T$.

Step 2. Dot with $x_B^T = (0, 1, 1)$, giving $0 + f_{21} + f_{23} + f_{31} + f_{33}$.

Step 3. Set the scalar to zero: $\boxed{f_{21} + f_{23} + f_{31} + f_{33} = 0}$.

Step 4. In row notation, $\boxed{(\mathbf{f}_2^T + \mathbf{f}_3^T) x_A = 0}$; this is one scalar epipolar incidence constraint.

Fall 2025 Quiz 2 — Three-view epipolar geometry **99/100**

Source: `cs532f25_quiz_2_solution.pdf`
Lecture match: Lecture 4, slides 9–21

Question 1. Three images A, B, C observe a point. The pair $x_A \leftrightarrow x_B$ is known, but its image location x_C is unknown. Given fundamental matrices F_{AC} and F_{BC} (which map A/B points to lines in C), determine x_C.

Step 1. Map x_A to a line in C: $\boxed{\ell_C^{(A)} = F_{AC} x_A}$. Any correct x_C must satisfy $x_C^T \ell_C^{(A)} = 0$.

Step 2. Map x_B independently to a second line in C: $\boxed{\ell_C^{(B)} = F_{BC} x_B}$. The same x_C must satisfy $x_C^T \ell_C^{(B)} = 0$.

Step 3. In the projective plane, the intersection of two lines is their cross product:

$$\boxed{x_C \sim \ell_C^{(A)} \times \ell_C^{(B)}}.$$

Step 4. Normalize $x_C = (q_1, q_2, q_3)^T$ to $(q_1/q_3, q_2/q_3)$ if $q_3 \neq 0$. Check both incidence products equal zero. If the two lines coincide, they do not determine a unique point; with noisy lines use an appropriate least-squares estimate.

Question 2. Given $x' = (1, 2, 1)^T$ and $F = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$, propose an epipolar-consistent point x.

Step 1. Start from $\boxed{x'^T F x = 0}$. To obtain the possible locations of x, form the line $\ell = F^T x'$ in the *unprimed* image.

Step 2. Multiply explicitly:

$$F^T = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \quad F^T x' = \begin{bmatrix} 1+1 \\ 2+2 \\ 1+4+1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$$

Step 3. Write line incidence for $x = (u, v, 1)^T$: $2u + 4v + 6 = 0$, or $\boxed{u + 2v + 3 = 0}$.

Step 4. Choose a convenient solution, e.g. $v = 0 \Rightarrow u = -3$. Thus $\boxed{x = (-3, 0, 1)^T}$. Alternatively $u = -1 \Rightarrow v = -1$.

Step 5. Verify the proposed answer: $Fx = (-2, 2, -2)^T$, hence $x'^T F x = (-2) + 2(2) + (-2) = 0$.

Exam check / qualification. The uploaded solution writes $x'^T F = x$; that is dimensionally/conceptually inaccurate. $x'^T F$ is a *line row-vector*, not a specific corresponding image point. There are infinitely many epipolar-consistent points on this line. Note that a true fundamental matrix must have rank 2; the numerical matrix given in the quiz is nonsingular, but its scalar incidence equation remains solvable as requested.

Fall 2025 Quiz 3 — Stereo depth sensitivity **98/100**

Source: `cs532f25_quiz_3_solution.pdf`
Lecture match: Lecture 4, slides 63–69

Question 1. Rectified stereo cameras have baseline $b = 10$ cm, common focal length $f = 50$ pixels, and true disparity $d^* = 25$ pixels. Determine the depth error caused by a 1-pixel disparity mistake.

Given / relevant facts: Use $Z = bf/d$. The problem does not say whether disparity is over- or underestimated, so compute both; the supplied key uses a first-order approximate magnitude.

Step 1. Keep physical units consistent: $bf = (10\,\text{cm})(50\,\text{px}) = 500\,\text{cm px}$. True depth:

$$Z^* = 500/25 = \boxed{20\,\text{cm}}.$$

Step 2. If disparity is underestimated by one pixel, $\hat{d} = 24$:

$$\hat{Z}_- = 500/24 = 20.83333\,\text{cm}, \quad \boxed{\Delta Z_- = +0.83333\,\text{cm}}.$$

Step 3. If disparity is overestimated by one pixel, $\hat{d} = 26$:

$$\hat{Z}_+ = 500/26 = 19.23077\,\text{cm}, \quad \boxed{\Delta Z_+ = -0.76923\,\text{cm}}.$$

Step 4. To obtain the first-order approximation, differentiate $Z(d) = bf/d$:

$$\frac{dZ}{dd} = -\frac{bf}{d^2}, \quad \Delta Z \approx -\frac{500}{25^2}\Delta d = -0.8\Delta d\ \text{cm}.$$

Thus the *approximate magnitude* is $\boxed{0.8\,\text{cm}}$ for $|\Delta d| = 1$.

Exam check / qualification. The exact errors are asymmetric ($+0.8333$ and -0.7692 cm), whereas the linearized approximation gives ± 0.8 cm. The source key reports the approximation, not both exact cases.

Question 2. For the same setup, restrict the searched 3D depth range to between 1 m and 5 m. What disparity interval corresponds to this?

Step 1. Convert baseline: $b = 10\,\text{cm} = 0.1\,\text{m}$. Then $bf = 0.1(50) = 5\,\text{m px}$.

Step 2. Invert the depth equation: $d = bf/Z$. Larger depth implies smaller disparity.

Step 3. At farthest depth $Z_{\max} = 5$ m: $d_{\min} = 5/5 = \boxed{1\,\text{px}}$.

Step 4. At nearest depth $Z_{\min} = 1$ m: $d_{\max} = 5/1 = \boxed{5\,\text{px}}$. Search $\boxed{d \in [1, 5]\,\text{px}}$.

Spring 2019 Quiz 5 — Exact stereo errors **98/100**

Source: `cs532s19_quiz5-2.pdf`
Lecture match: Lecture 4, slides 63–69

Problem 1(a–c). Rectified cameras have baseline $b = 10$ cm and focal length $f = 5$ pixels. True disparity is $d^* = 3$ pixels. (a) Find the depth error when disparity is underestimated by one pixel; (b) when overestimated by one pixel; (c) explain whether these errors remain constant at larger disparity.

Step 1. Write the depth-disparity law and insert the givens: $Z(d) = bf/d = (10)(5)/d = 50/d$ cm.

Step 2. True depth is $Z^* = 50/3 = \boxed{16\tfrac{2}{3}\,\text{cm}} \approx 16.6667$ cm.

Step 3. Underestimate disparity: $\hat{d} = 3 - 1 = 2$ px; $\hat{Z} = 50/2 = 25$ cm, so

$$\boxed{\Delta Z = \hat{Z} - Z^* = 25 - 50/3 = +25/3\,\text{cm} \approx +8.3333\,\text{cm}}.$$

Positive means the object appears *farther*.

Step 4. Overestimate disparity: $\hat{d} = 3 + 1 = 4$ px; $\hat{Z} = 50/4 = 12.5$ cm. Thus

$$\boxed{\Delta Z = 12.5 - 50/3 = -25/6\,\text{cm} \approx -4.1667\,\text{cm}}.$$

Negative means it appears *closer*.

Step 5. No: $Z(d) = bf/d$ is nonlinear. The exact error for $+1$ px is $Z(d+1) - Z(d) = -bf/[d(d+1)]$; for -1 px it is $Z(d-1) - Z(d) = +bf/[d(d-1)]$ for $d > 1$. Both magnitudes diminish at larger d.

Step 6. The derivative $dZ/dd = -bf/d^2$ gives the local approximation $|\Delta Z| \approx bf|\Delta d|/d^2$. Because $1/d^2$ decreases as d grows, disparity errors cause smaller depth errors for nearer points.

Spring 2020 Quiz 5 — Repeat disparity test **98/100**

Source: `cs532s20_quiz5.pdf`
Lecture match: Lecture 4, slides 63–69

Near duplicate: The uploaded quiz duplicates rank 5, including all three subquestions.

Problem 1(a–c). Rectified stereo cameras: $b = 10$ cm, $f = 5$ pixels, true $d^* = 3$ pixels. Determine (a) error for $\hat{d} = 2$, (b) error for $\hat{d} = 4$, and (c) how errors behave as disparity increases.

Step 1. Write $Z(d) = bf/d$, using $bf = (10\,\text{cm})(5\,\text{px}) = 50\,\text{cm px}$.

Step 2. True depth: $Z^* = 50/3 = 16.6667$ cm.

Step 3. Underestimate disparity to $\hat{d} = 2$: $\hat{Z} = 50/2 = 25$ cm; signed error $\hat{Z} - Z^* = 25 - 16.6667 = \boxed{+8.3333\,\text{cm}}$.

Step 4. Overestimate disparity to $\hat{d} = 4$: $\hat{Z} = 50/4 = 12.5$ cm; signed error $\hat{Z} - Z^* = 12.5 - 16.6667 = \boxed{-4.1667\,\text{cm}}$.

Step 5. Explain why the two numbers differ: Z is proportional to $1/d$, not linear in d; changes of $+1$ and -1 around the same d need not have equal effects.

Step 6. Generalize: $\frac{dZ}{dd} = -bf/d^2$, so a fixed disparity error typically matters less at larger disparities. For exact $+1$ px, $\Delta Z = -bf/[d(d+1)]$.

Lecture 5 — Corners and Feature Tracking

Lecture study note. Coverage gaps: numerical Harris calculations, second-moment eigenvalues, Lucas–Kanade normal equations, descriptors and matching metrics. See supplemental examples.

Relevance scores in this section are from the original quiz-matching rubric, not solution grades.

Fall 2025 Quiz 4 — KLT with auto-exposure **95/100**

Source: `cs532f25_quiz_4_solution.pdf`
Lecture match: Lecture 5, slides 39–50

Question 1. You track features over a long monocular video with automatic exposure that changes brightness from frame to frame. Propose modifications to the KLT brightness-constancy formulation, explaining why they improve robustness without invalidating the local linearization.

Step 1. State original KLT assumption: $I_2(x+\mathbf{d}) \approx I_1(x)$, with $\mathbf{d} = (u,v)^T$. Linearizing gives $I_x u + I_y v + I_t \approx 0$.

Step 2. Model auto-exposure as a frame-dependent affine brightness change: $I_2(x+\mathbf{d}) \approx aI_1(x) + b$, where $a > 0$ rescales contrast and b changes brightness. This violates raw brightness constancy even if motion is correct.

Step 3. For each local patch W, calculate its mean $\mu = (1/|W|)\sum_W I$ and standard deviation $\sigma = \sqrt{(1/|W|)\sum_W (I-\mu)^2}$. Define normalized intensity $\boxed{\widetilde{I}(x) = (I(x)-\mu)/(\sigma+\varepsilon)}$, with a small ε preventing division by zero.

Step 4. Show why the correction works when $\varepsilon \approx 0$: if $I_2 = aI_1 + b$ and $a > 0$, then $\mu_2 = a\mu_1 + b$ and $\sigma_2 = a\sigma_1$. Therefore

$$\widetilde{I}_2 = \frac{aI_1 + b - (a\mu_1 + b)}{a\sigma_1} = \frac{I_1 - \mu_1}{\sigma_1} = \widetilde{I}_1.$$

Step 5. Now apply the usual local Taylor expansion to *normalized* patches: $\widetilde{I}_2(x+\mathbf{d}) \approx \widetilde{I}_2(x) + \nabla\widetilde{I}_2(x)^T\mathbf{d}$. The tracking equations retain the Lucas–Kanade/KLT least-squares form but use normalized derivatives.

Step 6. Optional alternative: estimate exposure nuisance parameters jointly by minimizing $\sum_W [I_2(x+\mathbf{d}) - aI_1(x) - b]^2$; linearize the residual with respect to small increments in u, v, a, b.

Exam check / qualification. The uploaded key specifically proposes patch mean/std normalization. This method assumes patches have nonzero texture and a roughly affine photometric change. Saturation, shadows and spatially nonuniform lighting can still break the model.

Spring 2019 Quiz 6 — Feature trackability **91/100**

Source: `cs532s19_quiz6-2.pdf`
Lecture match: Lecture 5, slides 11–30 and 39–50

Problem 1(a–d). A binary image contains three black lines meeting at one vertex v: horizontal ℓ_h, vertical ℓ_v, and a 45° line ℓ_s. There is also one isolated black pixel $p = (x, y, 1)$ surrounded by white pixels. (a) Express intersection vertex coordinates in terms of the horizontal/vertical lines. (b) Is the common vertex good to track? (c) What about an arbitrary point along the slanted edge with both I_x and I_y large? (d) Is the isolated black pixel better than a neighboring white pixel?

Step 1. Represent a homogeneous line as $\ell = (a, b, c)^T$, with line equation $ax + by + cw = 0$. Two noncoincident lines intersect at their cross product, giving

$$\boxed{v \sim \ell_h \times \ell_v}.$$

Verify that the diagonal line also passes through the vertex: $\ell_s^T v = 0$.

Step 2. Use the Harris/KLT structure tensor $M = \sum_W w \nabla I \nabla I^T$. A corner-like intersection provides intensity gradients in multiple directions, making *both* eigenvalues sufficiently large. So yes, the vertex is generally good to track if patch size/contrast resolve the junction.

Step 3. For a straight diagonal edge, gradients may have *both coordinate components* $I_x \neq 0$, $I_y \neq 0$, but every local gradient is approximately parallel to the same edge normal g. Then $M \approx (\sum w\alpha_i^2) g g^T$, a rank-1 matrix with one near-zero eigenvalue. Thus no, not reliably trackable along the edge: the aperture problem remains.

Step 4. For an isolated dot, local intensity changes in both x and y directions; the dot as a *patch* is distinctive. However a nearby white pixel whose tracking *window includes the same dot* sees essentially the same local gradients and is not necessarily worse. So feature quality depends on the patch, not the center pixel color.

Step 5. If the tracking window around the neighboring white pixel *misses* the dot completely, then it is flat and poor. The correct answer requires specifying the window and its gradients.

Lecture 6 — Structure from Motion

Lecture study note. Coverage gaps: SVD factorization, triangulation, reprojection error, bundle adjustment, Gauss–Newton and Levenberg–Marquardt. See supplemental examples.

Relevance scores in this section are from the original quiz-matching rubric, not solution grades.

Fall 2025 Quiz 5 — SfM reconstruction upgrades **92/100**

Source: `cs532f25_quiz_5_solution-1.pdf`
Lecture match: Lecture 6, slides 11–19 and 25–45

Question 1, Scenario A. You photograph an office facade with parallel horizontal bands and parallel pedestrian crossing lines on the road. Is this geometry sufficient to upgrade an entire-scene projective reconstruction? To what type? What extra information is needed?

Step 1. Recall reconstruction hierarchy: projective transformations preserve incidence; affine preserves parallelism; similarity preserves angles and length ratios; Euclidean/metric reconstruction preserves true lengths when scale is fixed.

Step 2. Each known family of 3D parallel lines gives a vanishing/ideal point in projective reconstruction. Horizontal facade bands provide one direction; road crossing lines provide a second.

Step 3. To identify the plane at infinity π_∞ (a 3D plane in projective space), three independent noncollinear ideal points are generally needed. Two ideal points alone do not specify this plane.

Step 4. Therefore the two supplied families are insufficient for a unique entire-scene affine upgrade under the simple count used by the quiz. A third independent family, such as vertical facade edges, may provide the missing ideal point; with suitable calibrated correspondences it constrains the affine plane.

Step 5. To reach metric/similarity reconstruction, also require enough Euclidean information, such as camera calibration, orthogonality/angular or absolute-conic constraints with adequate geometry. One known physical length then fixes absolute scale.

Question 1, Scenario B. You film a planar kitchen floor tiled with regular square tiles. Can this geometry upgrade the reconstruction, and what else is required for metric scale?

Step 1. A square tile grid contains two line directions with known 3D parallelism, and their intersection angle is 90°. Equal-length squares also supply metric relations *within the floor plane*.

Step 2. The square grid directly permits a *metric rectification of the floor plane up to scale.* An *entire-scene similarity reconstruction (metric up to scale)* additionally requires sufficient camera-calibration and nondegenerate 3D constraints; diverse views of a single plane alone are not a general guarantee.

Step 3. If tile side length L is given (e.g. 30 cm), compare reconstructed tile dimension $\hat{L}$ with L and scale all translations/points by $s = L/\hat{L}$ to fix absolute units.

Step 4. Scope qualification: the known square grid alone, viewed only as a single plane without adequate camera/model constraints, does not in general uniquely determine full 3D scene metric geometry. State the extra assumption of sufficient viewpoints/calibration constraints when claiming an *entire-scene* upgrade.

Question 2. An image sequence has weak overlap: A overlaps B, B overlaps C, but A and C never share points, and so on. Is a single global factorization or a sequential SfM approach better?

Step 1. Form the tracking/measurement matrix mentally: many 3D features occur in only neighboring frames, leaving missing entries outside a narrow image-track visibility band.

Step 2. Classical complete-data factorization $D \approx MS$ presumes adequate common visibility. With widespread missing observations, naive dense SVD cannot be applied directly and estimates can become poorly conditioned.

Step 3. Choose sequential / incremental SfM: start with a strongly overlapping initial pair; estimate two-view geometry and poses, triangulate their tracks, add the next overlapping image via pose estimation, then triangulate new points.

Step 4. Periodically refine all visible cameras and points with bundle adjustment; robust matching handles outliers. This respects the chain-like visibility structure.

Exam check / qualification. The original answer key describes a similarity upgrade. The rigorous distinction is between rectifying the observed tile plane and uniquely upgrading arbitrary off-plane 3D structure; the latter needs extra camera/scene constraints.

Fall 2025 Quiz 6 — Large-scale Structure from Motion **76/100**

Source: `cs532f25_quiz_6_solution.pdf`
Lecture match: Lecture 6, slides 43–46; additional applied pipeline concepts

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Question 1. Given about 500,000 weakly overlapping photographs of Manhattan, choose incremental, global, hierarchical or hybrid SfM and defend your choice.

Step 1. Read the constraints: huge dataset, weak overlap, large repetitive urban textures (windows/facades), and significant computational burden from repeatedly refining poses and points.

Step 2. Incremental SfM grows a reconstruction camera by camera; it gives good local control but updating an enormous model can become expensive. Global SfM estimates camera motions jointly but bad pairwise matches in repetitive facades can contaminate the solution.

Step 3. Choose hierarchical SfM (a hierarchical–hybrid strategy is also defensible). Build an image-overlap graph and partition it into moderately sized, sufficiently connected image clusters.

Step 4. Run robust local two-view matching/geometric verification, initialize and triangulate points within each cluster, and perform local bundle adjustment.

Step 5. Merge neighboring cluster reconstructions using common tracks/cameras and an appropriate similarity/alignment transformation; reject inconsistent alignments. Perform regional/global adjustment once the merged model is stable.

Step 6. State why this choice matches the question: computation is localized; false repetitive-feature matches are easier to reject; partial overlaps need not be tracked in a single fully dense global factorization.

Exam check / qualification. The uploaded key uses an $O(N^3)$ assertion about pure incremental BA. That is not a universal complexity for sparse bundle adjustment, so the defensible argument is high computational cost that depends on graph structure, implementation and solver.

Question 2. The images are crowd-sourced and contaminated by "WTF" watermarks/text/frames that may produce false geometric matches. Suggest one simple modification to the RANSAC geometric verification step.

Step 1. Run RANSAC for each candidate image pair, estimating a valid geometric model and collecting its consensus inliers. A high inlier count *alone* may be deceptive when static overlays match each other.

Step 2. Measure the spatial distribution of inliers: compute the proportion near thin image-border strips, as watermark/frames often occur along borders.

Step 3. For those border inliers, calculate pixel-coordinate displacements $\delta_i = x_i' - x_i$ after dehomogenization (and accounting for image size/crop if needed). Check whether most are approximately the same 2D translation with little parallax or 3D structure.

Step 4. If both conditions hold—large border concentration and near-translation consistency—*reject or downweight the model/image pair*, despite its inlier count. Optionally mask border strips and rerun robust estimation.

Step 5. To avoid false rejection of genuine pure translations, apply this rule as a *heuristic* together with spatial-coverage and scene-consistency checks, not as an infallible theorem.

Supplement: seven worked problems missing from the old quizzes

The next seven examples are instructor-style *practice problems written for this guide*, not questions copied from the uploaded quiz PDFs. They address the mathematical topics in the six lecture decks that the 18 historical quizzes leave underrepresented. Each starts from a fully stated illustrative input.

A1. Homography, DLT and SVD (L3, slides 34–45)

Illustrative problem. Find the homography mapping $(0,0)\mapsto(0,0)$, $(1,0)\mapsto(1/2,0)$, $(0,1)\mapsto(0,1)$, $(1,1)\mapsto(1/2,1/2)$. Explain where SVD would enter for noisy pairs.

Step 1. Write the unknown homography $H=[h_{ij}]$, with $h_{33}=1$ for this convenient example. Use $u=(h_{11}x+h_{12}y+h_{13})/(h_{31}x+h_{32}y+1)$ and $v=(h_{21}x+h_{22}y+h_{23})/(h_{31}x+h_{32}y+1)$.

Step 2. The origin maps to itself, so at $(x,y)=(0,0)$: $h_{13}=h_{23}=0$.

Step 3. At $(1,0)\to(1/2,0)$: $h_{11}/(h_{31}+1)=1/2$ and $h_{21}/(h_{31}+1)=0$. Therefore $2h_{11}=h_{31}+1$ and $h_{21}=0$.

Step 4. At $(0,1)\to(0,1)$: $h_{12}/(h_{32}+1)=0$ and $h_{22}/(h_{32}+1)=1$. Thus $h_{12}=0$ and $h_{22}=h_{32}+1$.

Step 5. At $(1,1)\to(1/2,1/2)$: $h_{11}/(h_{31}+h_{32}+1)=1/2$ and $h_{22}/(h_{31}+h_{32}+1)=1/2$. Compare with step 3 to get $h_{32}=0$, hence $h_{22}=1$. Then $h_{31}+1=2$, so $h_{31}=h_{11}=1$.

Step 6. Assemble and verify:

$$\boxed{H=\begin{bmatrix}1&0&0\\0&1&0\\1&0&1\end{bmatrix}}.$$

For $(x,y)=(2,1)$, $H(2,1,1)^T=(2,1,3)^T$, so the new image coordinates are $(2/3,1/3)$.

Step 7. In general, each match $(x,y)\to(u,v)$ yields two rows of $Ah=0$:

$$A_i=\begin{bmatrix}x&y&1&0&0&0&-ux&-uy&-u\\0&0&0&x&y&1&-vx&-vy&-v\end{bmatrix}.$$

With more than four noisy pairs, choose the *last right singular vector* of $A=U\Sigma V^T$ (smallest singular value) and reshape it to H. Coordinate normalization before DLT improves conditioning: $H=T'^{-1}\widetilde{H}T$.

Exam check / qualification. Fixing $h_{33}=1$ is only valid if the actual h_{33} is nonzero; the homogeneous SVD formulation avoids relying on that choice.

A2. Harris response from gradients (L5, slides 11–30)

Illustrative problem. A small patch contains two gradient vectors $(2,0)^T$ and $(0,3)^T$, equally weighted. Compute its second-moment matrix, eigenvalues, and Harris score for $k=0.04$. Classify it.

Step 1. Construct the unweighted second-moment matrix as the sum of gradient outer products: $M=\sum_i g_i g_i^T$.

Step 2. Calculate each term: $g_1g_1^T=\begin{bmatrix}4&0\\0&0\end{bmatrix}$ and $g_2g_2^T=\begin{bmatrix}0&0\\0&9\end{bmatrix}$. Therefore $\boxed{M=\operatorname{diag}(4,9)}$.

Step 3. Because M is diagonal, its eigenvalues are $\lambda_1=4$ and $\lambda_2=9$; intensity changes in two independent directions.

Step 4. Compute determinant and trace: $\det M=4(9)=36$, $\operatorname{tr}M=4+9=13$.

Step 5. Harris response: $R=\det M-k(\operatorname{tr}M)^2=36-0.04(13)^2=36-6.76=\boxed{29.24}$.

Step 6. This positive score is *corner-like*; actual detector selection also requires an image-dependent threshold and spatial nonmaximum suppression. If instead $M=\operatorname{diag}(20,0)$, the response would be $-0.04(20)^2=-16$ (edge-like and singular).

A3. Lucas–Kanade least-squares displacement (L5, slides 39–50)

Illustrative problem. Two pixels in one small tracked patch have spatial gradients $(I_x, I_y) = (2, 0)$ and $(0, 3)$, with temporal derivatives $I_t = -4$ and -3, respectively. Find the shared displacement (u, v).

Step 1. Start with linearized brightness constancy at each pixel: $I_x u + I_y v + I_t = 0$.

Step 2. Substitute pixel 1: $2u + 0v - 4 = 0 \Rightarrow 2u = 4$. Pixel 2: $0u + 3v - 3 = 0 \Rightarrow 3v = 3$.

Step 3. Write matrix form: $Ad = b$, with $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $d = (u, v)^T$, $b = (4, 3)^T$.

Step 4. Compute the normal equations: $A^T A = \text{diag}(4, 9)$ and $A^T b = (8, 9)^T$, so $\begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} (u, v)^T = (8, 9)^T$.

Step 5. Solve: $\boxed{u = 2,\ v = 1}$ pixels (under small-motion/local-translation assumptions). Residuals in both equations are zero.

Step 6. Notice $A^T A = M$, the same second-moment matrix as in Harris. When both eigenvalues are positive and not too small, the optical-flow estimate is better determined; on an edge one eigenvalue tends toward zero and the aperture problem returns.

A4. Essential matrix and cheirality (L4, slides 22–41)

Illustrative problem. Two calibrated cameras have relative rotation $R = I$ and translation $t = (1, 0, 0)^T$ in compatible normalized units. Form E, identify its rank and singular values, and explain why pose decomposition is not unique without a positive-depth test.

Step 1. The essential matrix is $\boxed{E = [t]_\times R}$, with cross-product matrix $[t]_\times = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}$.

Step 2. Insert $t = (1, 0, 0)^T$ and $R = I$:

$$\boxed{E = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}}.$$

Step 3. Compute $E^T E = \text{diag}(0, 1, 1)$. Thus its singular values are $(1, 1, 0)$ (in some order), so $\text{rank}(E) = 2$ with two equal nonzero singular values, as required.

Step 4. For calibrated normalized image coordinates $\hat{x}, \hat{x}'$, corresponding points satisfy $\hat{x}'^T E \hat{x} = 0$. If pixel intrinsics are known, $E = K_R^T F K_L$.

Step 5. Decomposing a general essential matrix via SVD gives two rotation candidates R_1, R_2 and a translation *direction* with two signs, i.e. four candidates $(R_1, +t)$, $(R_1, -t)$, $(R_2, +t)$, $(R_2, -t)$.

Step 6. Triangulate one or more matched 3D points under each candidate. Keep the one for which the points have positive camera-frame Z in *both* views (cheirality). The magnitude of t remains unobservable without external scale.

A5. Linear triangulation with two projection matrices (L6, slides 48–52)

Illustrative problem. Rectified cameras have $f = 800$ px and baseline $b = 0.2$ m. Let $P_L = \begin{bmatrix} 800 & 0 & 0 & 0 \\ 0 & 800 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ and $P_R = \begin{bmatrix} 800 & 0 & 0 & -160 \\ 0 & 800 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. Observed pixels are $(260, 120)$ and $(220, 120)$. Reconstruct $X = (X, Y, Z)$.

Step 1. For each point (u, v) in a camera with matrix rows p_1^T, p_2^T, p_3^T, the two homogeneous equations are $(up_3^T - p_1^T)\widetilde{X} = 0$ and $(vp_3^T - p_2^T)\widetilde{X} = 0$, with $\widetilde{X} = (X, Y, Z, 1)^T$.

Step 2. Stack all four rows from the two cameras (using the negative of these rows, which is equivalent):

$$\boxed{A = \begin{bmatrix} -800 & 0 & 260 & 0 \\ 0 & -800 & 120 & 0 \\ -800 & 0 & 220 & 160 \\ 0 & -800 & 120 & 0 \end{bmatrix}}.$$

Step 3. Row 1 gives $X = (260/800)Z$ and row 2 gives $Y = (120/800)Z$. Row 3 gives $-800X + 220Z + 160 = 0$.

Step 4. Substitute X: $-260Z + 220Z + 160 = 0 \Rightarrow Z = 4$ m. Then $X = 260(4)/800 = 1.3$ m, $Y = 120(4)/800 = 0.6$ m.

Step 5. Answer $\boxed{(X, Y, Z) = (1.3, 0.6, 4) \text{ m}}$. Check by projecting into each camera: $(800X/Z, 800Y/Z) = (260, 120)$; $(800(X - 0.2)/Z, 800Y/Z) = (220, 120)$.

Step 6. With noisy measurements the four rays/equations are inconsistent. Use SVD $A = U\Sigma V^T$; take the last right singular vector as homogeneous $\widetilde{X}$, then divide by its fourth coordinate. Refine if needed by minimizing *reprojection* error rather than algebraic residual.

A6. Gauss–Newton and reprojection error (L6, slides 46 and 62–70)

Illustrative problem. For calibrated rectified stereo with $u_L = 260$, $bf = 160$ pixel-meters, and an observed right pixel $u_R^{obs} = 219$, refine an initial depth $Z_0 = 4$ m by one Gauss–Newton iteration.

Step 1. Predicted right pixel from stereo geometry is $\widehat{u}_R(Z) = 260 - 160/Z$. Define signed residual $r(Z) = \widehat{u}_R(Z) - 219 = 41 - 160/Z$. Minimize $E(Z) = r(Z)^2$.

Step 2. Evaluate initial prediction: $\widehat{u}_R(4) = 260 - 160/4 = 220$, hence $r(4) = 1$ pixel and $E(4) = 1$ squared pixel.

Step 3. Differentiate: $J(Z) = dr/dZ = 160/Z^2$, so $J(4) = 160/16 = 10$ pixels/meter.

Step 4. Gauss–Newton normal equation is $(J^T J)\Delta Z = -J^T r$. In this one-variable case $10^2 \Delta Z = -(10)(1)$, giving $\Delta Z = -0.1$ m.

Step 5. Update $\boxed{Z_1 = Z_0 + \Delta Z = 3.9 \text{ m}}$. New pixel prediction is $260 - 160/3.9 \approx 218.97436$, residual -0.02564, error $E(Z_1) \approx 0.000657$.

Step 6. The exact zero-residual solution is $Z^* = 160/41 \approx 3.902439$ m. Full bundle adjustment extends this to many camera pose and 3D-point variables: $\min_\theta \sum \|x^{obs} - \pi(P(\theta)X(\theta))\|^2$. LM adds damping: $(J^T J + \lambda I)\Delta\theta = -J^T r$.

Exam check / qualification. This is a one-variable reprojection refinement, not full joint bundle adjustment. The purpose is to demonstrate a complete numerical optimization update before scaling to the sparse multivariable system.

A7. Affine structure-from-motion factorization (L6, slides 25–37)

Illustrative problem. Four 3D points are $(0,0,0)$, $(2,0,0)$, $(0,2,0)$ and $(0,0,2)$. The first orthographic camera records (X, Y) and the second records (Y, Z). Build the centered observation matrix and explain its rank and SVD factorization.

Step 1. In view 1, image x coordinates are $(0, 2, 0, 0)$ with mean 0.5, and y coordinates are $(0, 0, 2, 0)$ with mean 0.5. Centered rows: $(-.5, 1.5, -.5, -.5)$ and $(-.5, -.5, 1.5, -.5)$.

Step 2. In view 2, image coordinates are (Y, Z). Their centered rows are $(-.5, -.5, 1.5, -.5)$ and $(-.5, -.5, -.5, 1.5)$.

Step 3. Stack views to obtain the $2m \times n = 4 \times 4$ matrix:

$$\boxed{D = \begin{bmatrix} -.5 & 1.5 & -.5 & -.5 \\ -.5 & -.5 & 1.5 & -.5 \\ -.5 & -.5 & 1.5 & -.5 \\ -.5 & -.5 & -.5 & 1.5 \end{bmatrix}}.$$

Step 4. Rows 2 and 3 are identical, so rank ≤ 3. Rows 1,2,4 are linearly independent; therefore $\boxed{\text{rank}(D) = 3}$. This illustrates the centered affine/orthographic SfM rank-three condition.

Step 5. Perform SVD $D = U\Sigma V^T$, retain the top three singular terms, and form $\widetilde{M} = U_3\Sigma_3^{1/2}$, $\widetilde{S} = \Sigma_3^{1/2}V_3^T$. Then $D = \widetilde{M}\widetilde{S}$ in this noise-free example.

Step 6. The factorization is not unique: for any invertible $Q \in \mathbb{R}^{3\times3}$, $(\widetilde{M}Q)(Q^{-1}\widetilde{S}) = D$. Metric constraints and camera-model assumptions are needed to upgrade from arbitrary affine coordinates. With missing feature observations, naive dense SVD also needs a missing-data strategy.